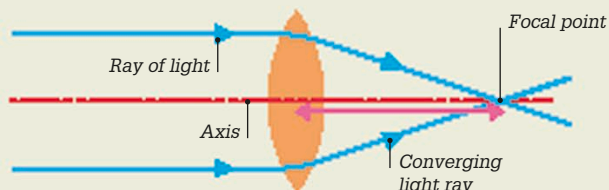


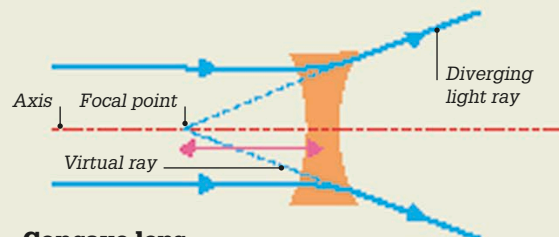
Bending light

Scientists study the nature of light and also how it can be manipulated. The study of how lenses work is an example of this.



Convex lens

A convex lens bulges in the middle. When parallel rays of light pass through it, they bend inward (converge), and meet at the focal point just beyond the lens.



Concave lens

A concave lens is hollow in the middle. When parallel rays of light pass through it, they spread out (diverge), and appear to come from a focal point in front of the lens.

Optical illusion

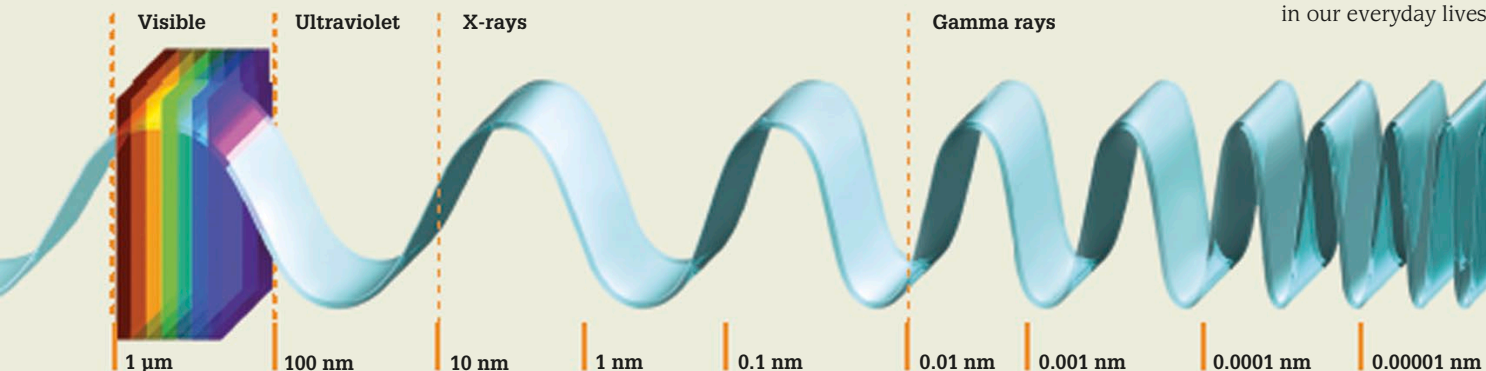
Our eyes can be tricked into seeing things that are not there, with the help of mirrors and reflections. The theatrical device Pepper's Ghost (named after one of its inventors) uses a large piece of angled glass to bounce the image of a ghostly figure on to the stage.



Pepper's Ghost fools a theater audience in the 19th century

Electromagnetic waves

Visible light forms a tiny part of the entire range (spectrum) of electromagnetic waves (below) that are traveling through space at extremely high speed, and were first described in the 19th century. The different types of waves have individual uses and functions in our everyday lives.



Visible light

Our eyes are able to see color within this range of wavelengths. Some animals can see beyond this range.



X-rays

X-rays can be used to see inside our bodies. They penetrate the soft tissue but are blocked by bones and teeth.

1704

Sir Isaac Newton published *Opticks*, in which he opposed Huygens's theory. He described light as a stream of tiny particles (corpuscles) traveling at huge speed in a straight line.

1816

French physicist Augustin-Jean Fresnel carried out a series of experiments that supported the wave theory of light.



Augustin-Jean Fresnel

1864

Scottish mathematician James Clerk Maxwell showed that light is an electromagnetic wave. He went on to predict the existence of other forms of electromagnetic waves.

1905

Physicist Albert Einstein united wave and particle theory by proposing that light is made up of tiny packets of energy called photons that also behave as waves.